



Vendor · Canonical Shape Directory

Reference · v8 · case-encoded across 4 dims

The 5-segment vendor identity — locked grammar. PAY_SEQ × SELLS × OPM × PMS × IDENTITY. All four architectural dimensions are case-encoded (uppercase primary + optional lowercase alt). Brand scope is still computed from the RM ring at query time.

SECTION 1

The 5-segment vendor identity

The vendor canonical code carries five locked dimensions: **PAY_SEQ** (cash-flow direction), **SELLS** (what they stock — RM scope), **OPM** (how we order), **PMS** (which payment methods they accept), and **IDENTITY** (unique abbreviation). Brand scope is still a **computed view** derived from the RM ring at query time.

v8 doctrine fix: all four architectural dimensions follow the same case-encoded pattern — **uppercase letter = primary rail/direction**, **lowercase letter = also possible (alternate)**, **letter absent = impossible**. The grammar fully encodes possibility from impossibility — no information is thrown away. Every dimension is enumerated; the constraint set rejects impossible combinations.

TOTAL VENDORS

—

live in editor

PAY-FIRST PRIMARY

—

Pf + Pfr

RECEIVE-FIRST PRIMARY

—

Rf + Rfp

LINKED VS ORPHAN

—

RM-referenced vendors

[➤ Vendor Editor](#)

[➤ RM Editor](#)

[➤ RM Shape Directory](#)

[➤ COA Doctrine](#)

[↓ Live Census](#)

SECTION 2

Code Anatomy — 5 segments, left to right

{PAY_SEQ} - {SELLS} - {OPM} - {PMS} - {IDENTITY}

SELLS

POSITION 2 · 1-2 CHARS · WHAT THEY STOCK (RM SCOPE)

L = loose only · B = branded only · Lb = loose primary + branded alt · Bl = branded primary + loose alt.
Flat LB retired in v8.

4 values · L · B · Lb · Bl

PMS

POSITION 4 · 1-2 CHARS · PAYMENT-METHOD SUPPORT

C = cash only · **B** = bank only · **Cb** = cash primary + bank alt · **Bc** = bank primary + cash alt.

4 values · C · B · Cb · Bc

IDENTITY

POSITION 5 · 3-10 CHARS UPPERCASE

Unique abbreviation drawn from the vendor name. Auto-derived (consonants vowel-stripped); owner-overrideable. Globally unique across all vendors regardless of the four leading dimensions.

PRABHU · ZEPTO · SHARIFF · HYPERPURE · WALKIN

Reading order: PAY_SEQ (cash direction) → SELLS (what they stock) → OPM (how we order) → PMS (how we pay) → IDENTITY (who they are). Five locked dimensions, all enumerated. **Universal case convention:** uppercase letter = primary, lowercase letter = also possible, letter absent = impossible. Brand scope still computed from the RM ring upstream.

SECTION 3

PAY_SEQ directory Position 1 · 2-3 chars

Cash-flow direction. v8: case-encoded primary + alt — same pattern that runs across SELLS, OPM, and PMS.

Letter table

CODE	MEANING	REAL EXAMPLES
Pf	Pay-first only. Money always leaves our account before goods/service arrives. Pure quick commerce, strict prepaid.	Pf-B-A-B-ZEPT0 · Pf-L-M-C-WALKIN
Rf	Receive-first only. Goods/service always arrives before money goes out. Pure daily milk, post-paid utilities, strict credit-term vendors.	Rf-L-M-C-PRABHU · Rf-Lb-M-Cb-SHARIFF
Pfr	Pay-first primary + Receive-first alt. Default flow is pay-first, but receive-first is also possible (e.g. a quick-commerce vendor that occasionally ships COD).	Pfr-B-Am-Bc-EXAMPLE (hypothetical)
Rfp	Receive-first primary + Pay-first alt. Default flow is receive-first, but pay-first happens occasionally (e.g. a credit-term vendor where rare items require advance payment).	Rfp-Lb-Ma-Cb-SHARIFF (hypothetical)

Case-encoded combinations · primary/alt convention

When a vendor's cash-flow direction admits both flows, the canonical code uses three letters: leading **uppercase** = primary direction; the trailing f is a legibility suffix; trailing **lowercase** letter = alternate direction. The dropped information from a flat binary — which direction dominates — is preserved here.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
PAY_SEQ outside {Pf, Rf, Pfr, Rfp}	Enumerated set of legal codes. The 'f' suffix anchors the primary letter; Prf / Rpf are rejected.

FORBIDDEN

WHY CAUGHT

Pf + Rf as separate vendors for the same entity

Once a real vendor admits both flows, it should be Pfr or Rfp (primary specified) — never two records.

Why not a 6-letter V-shape here?

The richer V-shapes (V1a / V1b / V2 / V3 / V4 / V5) capture the precise cash-flow choreography of a single purchase event. Those live at the *action layer* and modulate per-purchase, per-supplier within a single vendor record. PAY_SEQ at the vendor level only captures the directional possibility space (which flows are possible, which is dominant) — that's enough for the entity ring; the action ring fills in the granularity.

SECTION 3.5

SELLS directory Position 2 · 1-2 chars

What the vendor stocks at the RM-purchase layer. v8: case-encoded primary + alt. Flat LB retired.

Letter table

CODE	MEANING	REAL EXAMPLES
L	Loose only. Vendor only deals in unbranded RM by weight/volume — daily milk, vegetables, mutton, fish, walk-in emergency procurement.	Rf-L-M-C-PRABHU (loose milk)
B	Branded only. Vendor only stocks SKU-coded branded goods — quick commerce apps, B2B platforms, dedicated brand suppliers.	Pf-B-A-B-ZEPTO · Rf-B-A-Bc-HYPERPURE
Lb	Loose primary + Branded alt. Mixed shop where loose dominates (by volume/value) but branded packs are also stocked as a secondary line.	Rf-Lb-M-Cb-SHARIFF (loose-led departmental)
BL	Branded primary + Loose alt. Mixed shop where branded SKUs dominate but some loose staples also stocked.	Rf-BL-M-Cb-NILDIST (hypothetical brand-led distributor)

Case-encoded combinations · primary/alt convention

When a vendor stocks both loose and branded, the canonical code uses two letters: **uppercase** = primary stock type (the bulk of what they sell); **lowercase** = alternate (the secondary line). The information that was discarded by the flat LB alphabet is preserved here.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
SELLS = LB (flat alphabetical)	v8 rejects flat ordering — owner must declare a primary letter. Use Lb or Bl.
SELLS empty	Every RM-supplying vendor stocks at least one of {L, B}. Empty SELLs would be an unsupplied vendor — not a valid entity-ring state.
Lowercase first (lB) / both upper (LB) / duplicates (Ll)	Same constraints as PMS. Primary is uppercase; alt is lowercase; each letter at most once.
Letter outside {L, B}	The two-class enumeration is exhaustive in RM-purchase scope. New letters S/C reserved for the bill/expense scope.

Future SELLs extension

When the bill/expense scope opens (rent, utilities, services, capex), the SELLs alphabet extends to include S (services) and C (capex). New combinations like Sb, Cl, S become valid points in the same case-encoded space — **no migration required**. The constraint set widens; the existing {L, B, Lb, Bl} rows stay valid.

SECTION 3.6

OPM directory Position 3 · 1-2 chars

Order-Placement Method — how an order with this vendor is initiated. v8: case-encoded primary + alt. Same pattern as SELLs / PMS.

Letter table

CODE	MEANING	REAL EXAMPLES
M	Manual only. Ordering always routes through a human channel — phone call, WhatsApp, walk-in, or a fixed daily/weekly delivery route. No app/API in the loop.	Rf-L-M-C-PRABHU (daily route) · Rf-Lb-M-Cb-SHARIFF (WA)
A	Automatable only. Ordering always goes via app, web portal, or API — quick commerce, B2B platforms, marketplace integrations.	Pf-B-A-B-ZEPT0 (app) · Rf-B-A-Bc-HYPERPURE (B2B)
Ma	Manual primary + Automatable alt. Default ordering is manual but the vendor also exposes an app/API as a backup channel.	Rf-Lb-Ma-Cb-EXAMPLE (hypothetical hybrid)

CODE	MEANING	REAL EXAMPLES
Am	Automatable primary + Manual alt. Default ordering is via app/API but a human channel exists as fallback (call vendor when API down).	Pf-B-Am-B-EXAMPLEAPP (hypothetical)

Case-encoded combinations · primary/alt convention

When a vendor admits both ordering channels, the canonical code uses two letters: **uppercase** = primary method (the channel that handles most orders); **lowercase** = alternate (the backup channel). The primary letter is what gates whether agents emit orders by default.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
OPM outside {M, A, Ma, Am}	Same case-encoded pattern as SELLS / PMS.
Lowercase first (mA) / both upper (MA) / duplicates (Mm)	Primary uppercase, alt lowercase, each letter at most once.
Letter outside {M, A}	The two-method enumeration is exhaustive at the entity layer. New ordering channels would require a doctrine extension.

Why this dimension matters

OPM separates "needs a human to send WA" from "Cron Worker can place the order." The primary letter is the gate that decides whether an action becomes agent-emitted (A / Am) or staff-emitted (M / Ma) *by default*. The alt letter declares the fallback path is also available — useful for graceful degradation when the primary channel fails.

SECTION 3.7

PMS directory Position 4 · 1-2 chars

Payment-Method Support — which payment rails this vendor accepts. Two letters, case-encoded for hybrid primaries. Mirrors SOURCING Lb/Bᳵ and USAGE Pr/Rp.

Letter set · pure rails

CODE	MEANING	REAL EXAMPLES
C	Cash-only. Vendor doesn't accept bank rails — no UPI, no NEFT, no card. Settlement always physical.	Rf-L-M-C-PRABHU (small loose milk vendor)

CODE	MEANING	REAL EXAMPLES
B	Bank-only. Vendor doesn't accept cash — typical for app-based platforms, registered B2B suppliers, online subscriptions, GST-registered wholesalers. UPI / NEFT / card / cheque only.	Pf-B-A-B-ZEPT0

Case-encoded combinations · hybrid rails

When a vendor accepts both rails, the canonical code uses two letters: **uppercase** = primary (default) rail, **lowercase** = alternate. The information thrown away by a flat H letter — which rail is the default — is preserved here.

CODE	MEANING	REAL EXAMPLES
Cb	Hybrid · Cash primary. Vendor accepts both, but cash is the default rail; bank is the alternate. Common for small/local vendors who occasionally take UPI.	Rf-Lb-M-Cb-SHARIFF (departmental store)
Bc	Hybrid · Bank primary. Vendor accepts both, but bank is the default rail; cash is the alternate. B2B platforms / registered vendors with COD option.	Rf-B-A-Bc-HYPERPURE (B2B with cash possible)

Forbidden combinations · what the regex rejects

FORBIDDEN	WHY CAUGHT
PMS empty	Every vendor accepts at least one rail. Empty PMS would mean "we cannot pay this vendor at all" — not a valid operational state.
bC · cB	Lowercase-first reverses the primary/alternate convention. Primary letter must be uppercase.
Cc · Bb	Each letter (C, B) appears at most once. Duplicate is meaningless.
CB · BC	Both uppercase declares two primaries — only one rail can be primary at a time.
Letter outside {C, B}	The two-rail enumeration is exhaustive at the entity layer. New rails (e.g. crypto) would require a doctrine extension, not a free-form letter.

What PMS does NOT capture

PMS describes what the vendor *accepts* and *which rail is primary*, not what was used. The exact payment channel for a specific bill (UPI vs cash for a Cb vendor, NEFT vs UPI for a B vendor) is a property of the action — settled when the transaction happens. PMS only narrows the option space; the action picks one.

SECTION 4

IDENTITY directory Position 5 · 3-10 chars

A 3-10 character uppercase abbreviation that gives the vendor its identity within the global vendor namespace.

Length

3 to 10 characters, uppercase A-Z and 0-9 only. Codes need to stay scannable.

Source

Auto-derived from vendor name (consonants, vowels stripped, deduplicated). Owner override allowed.

Uniqueness

Globally unique across ALL vendors regardless of the four leading dimensions. Rf-L-M-C-PRABHU blocks Pf-B-A-B-PRABHU.

Case

Always uppercase. Editor and API normalise lowercase on save.

Sample derivations

VENDOR NAME	AUTO-DERIVED IDENTITY	NOTES
Prabhu Buffalo Milk Vendor	PRABHU	Lead noun retained
Zepto	ZEPTO	Brand name as-is
Shariff Departmental Store	SHARIFF	Lead noun
Rent — Hamza HKP	RENT	Functional handle

VENDOR NAME	AUTO-DERIVED IDENTITY	NOTES
BESCOM (Electricity)	ELECTRICITY	Function over brand
BSNL Internet	INTERNET	Function over brand

When two vendors play the same operational role (e.g. two electricity providers across outlets), use a discriminator suffix in IDENTITY — Pf-B-A-B-ELECTRICNCH, Pf-B-A-B-ELECTRICHE.

SECTION 5

Forbidden Combinations

Every anomaly the system rejects, in one place. The math literally cannot represent these.

FORBIDDEN	WHY CAUGHT	RESULT
PAY_SEQ outside {Pf, Rf, Pfr, Rfp}	Enumerated set; the 'f' suffix anchors the primary letter. Prf / Rpf rejected.	400 reject
SELLS = "LB" (flat)	v8 retired flat ordering — owner must declare a primary. Use Lb or B $\bar{L}$.	400 reject
SELLS outside {L, B, Lb, B $\bar{L}$ }	Same case-encoded pattern as PMS. S/C reserved for the bill/expense scope.	400 reject
OPM outside {M, A, Ma, Am}	Same case-encoded pattern.	400 reject
PMS outside {C, B, Cb, Bc}	Same case-encoded pattern.	400 reject
Any case-encoded segment with lowercase first / both upper / duplicate letter	Universal v8 constraint. Primary uppercase, alt lowercase, each letter at most once.	400 reject
IDENTITY empty	Identity required.	400 reject
IDENTITY < 3 chars	Codes need to stay readable.	400 reject
IDENTITY > 10 chars	Codes need to stay scannable.	400 reject
IDENTITY collision	Two vendors cannot share the same identity, even when the four leading dimensions differ.	409 conflict
Lowercase / non-alphanumeric	IDENTITY is uppercase A-Z and 0-9 only.	400 reject (auto-uppercased on input)

FORBIDDEN	WHY CAUGHT	RESULT
DELETE while RM-referenced	Cannot orphan an RM's supplier link silently — caller must remove from RM tree first.	409 conflict

SECTION 6

Live Vendor Census — All Vendors Today

Fetches live from `/api/vendors`. Each vendor shows its declared SELLS / OPM / PMS pills plus the brand scope computed from the RM ring. Click any code to open the editor.

Could not load live census from `/api/vendors` — Failed to fetch. The endpoint may need a valid PIN; this directory shares the editor's auth.

SECTION 7

Worked Examples — Reading codes left to right

Six codes spanning the v8 grammar: two pure-binary forms (Rf-L-M-C, Pf-B-A-B), the full-alternates form (Rfp-Lb-Ma-Cb), branded-primary hybrid (Rf-BL-M-Bc), and intermediate cases. Each decomposed segment by segment.

1 · Rf-L-M-C-PRABHU — Daily loose milk vendor

Rf-L-M-C-PRABHU | | | | L PRABHU — identity (loose buffalo milk vendor in Shivajinagar) | |
 | L C — Cash-only (small vendor, no bank rail at all) | | L M — Manual order (we
 WhatsApp him daily) | L L — Loose only (unbranded buffalo milk by litre)
 L Rf — Receive-first (delivers milk daily; settled weekly/monthly)

Reads as: "Receive-first, sells loose, manually ordered, cash-only vendor named PRABHU."
 Computed scope from RM ring = NCH (only NCH consumes buffalo milk).

2 · Pf-B-A-B-ZEPTO — Quick commerce platform

Pf-B-A-B-ZEPTO | | | | L ZEPTO — identity (the quick commerce platform) | | | L B —
 Bank-only (no cash settlement; in-app UPI/card payment) | | L A — Automatable (Cron
 Worker can place orders via app/API) | L B — Branded only (SKU-coded packs — Zepto
 doesn't deal in loose) L Pf — Pay-first (paid in-app, then 10-minute delivery)

Reads as: "Pay-first, branded only, automatable, bank-only vendor named ZEPTO." Note PMS
 position 4 carries a single uppercase B — distinct from SELLS at position 2, even though they
 share the letter shape. Computed scope = HN (cross-brand — both NCH and HE order from
 Zepto).

3 · Rfp-Lb-Ma-Cb-SHARIFF — Full alternates: every dim carries a primary + alt

Rfp-Lb-Ma-Cb-SHARIFF | | | | L SHARIFF – identity (Shivajinagar departmental store) | | |
 _____ Cb – Cash primary + Bank alt (cash is default; UPI accepted) | | _____ Ma –
 Manual primary + Auto alt (we WA mostly; vendor's app exists as backup) | _____ Lb –
 Loose primary + Branded alt (loose is the bulk; branded packs are secondary)
 _____ Rfp – Receive-first primary + Pay-first alt (settle monthly mostly; rare
 items prepaid)

Reads as: "Every dimension has both rails, but one is dominant: receive-first leads cash flow, loose leads stock, manual leads ordering, cash leads payment." This is the doctrine fix in action — the v7 grammar (Rf-LB-M-Cb-SHARIFF) couldn't say which direction / stock / method dominated; v8 puts that information into the code itself. The action layer reads coordinates and routes accordingly without a lookup.

4 · Rf-B-A-Bc-HYPERPURE — B2B platform with COD option

Rf-B-A-Bc-HYPERPURE | | | | L HYPERPURE – identity (Zomato's B2B sourcing platform) | | |
 _____ Bc – Hybrid · Bank primary + Cash alt (NEFT/UPI default; cash possible on COD
 orders) | | _____ A – Automatable (B2B portal supports API) | _____ B – Branded
 only (SKU-coded wholesale) _____ Rf – Receive-first (delivery + invoice; settle on
 credit terms)

Reads as: "Receive-first, branded only, automatable, bank-primary hybrid vendor named HYPERPURE." Distinct from ZEPTO on two dimensions: PAY_SEQ (Hyperpure invoices on credit, Zepto charges in-app upfront) and PMS (Bc vs B — Hyperpure has a fall-back cash option on COD).

5 · Pf-L-M-C-WALKIN — Emergency walk-in procurement

Pf-L-M-C-WALKIN | | | | L WALKIN – identity (functional handle for ad-hoc local buys) | | |
 _____ C – Cash-only (counter purchase, immediate cash; no bank rail) | | _____ M –
 Manual order (staff walks to local shop) | _____ L – Loose only (vegetables / mutton
 emergency top-up) _____ Pf – Pay-first (cash handed over before goods loaded)

Reads as: "Pay-first, loose only, manually ordered, cash-only vendor named WALKIN." A functional vendor record covering all ad-hoc emergency procurement — counts as one entity even though physical shops vary.

6 · Rf-BL-M-Bc-NILDIST — Branded primary + loose alt, bank-primary hybrid

Rf-BL-M-Bc-NILDIST | | | | L NILDIST – identity (hypothetical brand-led distributor) | | |
 _____ Bc – Bank primary + Cash alt (NEFT/UPI default; cash possible on rare COD) | |
 _____ M – Manual only (call/WA – no app yet) | _____ BL – Branded primary + Loose
 alt (brand-led wholesaler with some loose staples) _____ Rf – Receive-first
 (delivery + invoice; settle on credit terms)

Reads as: "Receive-first, branded-dominant with loose alt, manual ordering only, bank-primary hybrid vendor." Demonstrates the v8 grammar's reach: SHARIFF and NILDIST share PAY_SEQ (Rf) and OPM (M), but their SELLS primary (Lb vs BL) and PMS primary (Cb vs Bc) flip — a difference that v7's flat grammar couldn't represent. The day NILDIST exposes an API, a single PUT moves

the row to Rf-BL-Am-Bc-NILDIST; the canonical code re-keys; the action ring picks up the change with no migration.

SECTION 8

What stays out of the code

An obvious instinct would be to encode BRAND scope (NCH/HE/HN) directly in the vendor code: **Rf-L-M-C-NCH-PRABHU**. We deliberately don't — and the same discipline applies to several other tempting fields:

BRAND scope is derivable. It changes when the RM ring changes — without any vendor mutation. Encoding it statically would create sync drift the day NCH stops using PRABHU.

Per-transaction payment method. Whether a hybrid vendor was paid in UPI or cash for a specific bill is a property of the action, not the vendor. PMS narrows the option space; the action picks one.

Money source / L3 routing. Which bank account, which till, which UPI handle paid this vendor — that's a Layer-3 ledger concern, not a vendor identity concern.

Communication channel. Phone/WhatsApp/email is a property field. OPM captures whether ordering can be automated; the actual contact details are data_json.

Bank account / contact details. Pure properties — they live in data_json, never in the canonical code.

Five canonical dimensions; everything else is property or derived view.

In the static row

- vendor_code (Pf/Rf-SELLS-OPM-PMS-IDENTITY)
- pay_seq, sells, opm, pms, identity_abbr, vendor_name
- communication / location / payment / GST in data_json

Never in the static row

- BRAND scope (derivable from RM ring)
- Per-transaction payment method (action layer)
- Money source / L3 routing (ledger)
- Communication channel (property)
- Bank account / contact (property)
- V-shape (action layer)

Future SELLS dimension extension

The current SELLS alphabet is {L, B} with case-encoded values {L, B, Lb, Bl}, valid in RM-purchase scope only. When the bill/expense scope opens (rent, utilities, services, capex), the alphabet extends: S = services, C = capex. Combinations like S, Sb, Cl become valid — same case-encoded constraint pattern, just a wider letter set. **No migration required:** existing rows stay valid, the CHECK constraint widens, and new combinations slot in as new points in the same space.

SECTION 9

Connection to the RM ring + Action layer

The vendor ring is the **second entity ring**, sitting alongside the RM ring. Both feed downstream into the Action ring (purchases, productions, transfers).

Composition of an Action code (sketch)

`{ACTION}-{RM_CODE}-{VENDOR_CODE}-{V_SHAPE}-{TXN_ID}`

Examples:

PUR-NCH-AM-P-Lb-MLK-Rf-L-M-C-PRABHU-V1a-2026050801 ← daily milk receipt
PUR-HN-AM-P-Bl-BTR-Pf-B-A-B-ZEPT0-V3-2026050802 ← Amul butter via Zepto
PUR-HN-AM-P-Lb-VEG-Rfp-Lb-Ma-Cb-SHARIFF-V2-2026050803 ← veg + branded stuff

RM ring

Five segments — what was bought. Vendor refs live inside data_json at the L / B / I layers.

Vendor ring (this page)

Five segments — cash direction · what they sell · how we order · how they accept payment · who they are. Brand scope still computed via the RM ring.

Action ring

Composes RM + Vendor + V-shape into a transactional code that an Event can aggregate. Defined in COA Doctrine.

Two entity rings, one action ring, one event ring. Every action code is a coordinate; every event aggregates actions. The vendor ring stays minimal so the action ring stays composable.

